

FACTOR GROUPS:

Definition: Let $(G, \circ, e_G)$ be a group, we say that a subgroup $H \leq G$ is normal if for every $g \in G$ $gHg^{-1} = H$ i.e. $gH = Hg$. We write $H \trianglelefteq G$ to indicate it is normal in G .

Examples:

① Any subgroup $H \leq G$ if G is abelian is a normal subgroup.

$$gH = \{gh \mid h \in H\} \quad Hg = \{h \cdot g \mid h \in H\}$$

if G is abelian $gh = hg \quad \forall h \in H$ then $gH = Hg$.

② Let $G = GL_n(\mathbb{R})$ and $H = \{A \in GL_n(\mathbb{R}) \mid \det(A) = 1\}$

Then H is a normal subgroup.

(it is the kernel of $\det: GL_n(\mathbb{R}) \rightarrow (\mathbb{R}^*, \cdot, 1)$)
this subgroup is called the special linear group. ($SL_n(\mathbb{R})$)

③ A_n is normal in S_n , since it is the kernel

$$\text{sign}: S_n \rightarrow \{1, -1\}$$

In general:

Theorem: Let $(G, \circ_G, e_G)$ be a group and let

$\varphi: G \rightarrow (K, \circ_K, e_K)$ be a group homomorphism.

Let $\text{Ker}(\varphi) := \{g \in G \mid \varphi(g) = e_K\}$

then $\text{Ker}(\varphi)$ is a normal subgroup of G .

Proof: • $\text{Ker}(\varphi)$ is a subgroup:

• $e_G \in \text{Ker}(\varphi)$ since $\varphi(e_G) = e_K$.

• closed under $\circ_G$: Let $g_1, g_2 \in \text{Ker}(\varphi)$

we want to show $\varphi(g_1 \circ g_2) = e_K$ i.e. $g_1 \circ g_2 \in \text{Ker}(\varphi)$

since φ is an homomorphism $\varphi(g_1 \circ g_2) = \varphi(g_1) \circ_K \varphi(g_2)$
 $= e_K \circ_K e_K = e_K$.

• closed under inverses: Let $g \in \text{Ker}(\varphi)$ then
 $\varphi(g) = e_K$ since φ is an homomorphism

$$e_K = \varphi(e_G) = \varphi(g \circ g^{-1}) = \varphi(g) \circ_K \varphi(g^{-1})$$

||
 e_K

then $\varphi(g^{-1}) = e_K$.

normal: Let $g \in G$ we want to show that
 $g \text{Ker}(\varphi) g^{-1} = \text{Ker}(\varphi)$.

" $\subseteq$ " Let $h \in \text{Ker}(\varphi) \Rightarrow \varphi(h) = e_K$. Then $g \cdot h \cdot g^{-1}$
 $\in \text{Ker}(\varphi)$ as

$$\text{in particular } \psi(g \cdot h g^{-1}) = \psi(g) \cdot \psi(h) \cdot \psi(g^{-1}) \\ = \psi(g) \cdot e_K \cdot \psi(g)^{-1} = e_K.$$

$$\text{thus } g \text{Ker}(\psi) g^{-1} \subseteq \text{Ker}(\psi).$$

For the converse, we want to show that $\text{Ker}(\psi) \subseteq g \text{Ker}(\psi) g^{-1}$, let $h \in \text{Ker}(\psi)$

$$\text{let } h_1 = g^{-1} h g.$$

claim: $h_1 \in \text{Ker}(\psi)$

$$\psi(g^{-1} h g) = \psi(g)^{-1} \underbrace{\psi(h)}_{= e_K} \cdot \psi(g) = e_K$$

$$\Rightarrow h_1 \in \text{Ker}(\psi) \text{ then } \text{Ker}(\psi) \subseteq g \text{Ker}(\psi) g^{-1}.$$

Theorem Let H be a subgroup of G . then left coset multiplication

$$(aH) \cdot (bH) = ab \cdot H \text{ is well defined}$$

iff H is normal in G .

proof $\Leftarrow$ let $a_1 \in aH$ and $b_1 \in bH$

we want to show that $(a_1H)(b_1H) = a_1b_1H$
 $= abH.$

$$\cancel{a_1 = a h_1} \quad \cancel{b_1 = b \cdot h_2^*}$$

$$\cancel{a_1^{-1} \cdot a_1 = h_1} \quad \cancel{b_1^{-1} \cdot b_1 = h_2^*}$$

Let $h_1, h_2 \in H$ be such that $a_1 = a \cdot h_1$ and $b_1 = b \cdot h_2$.

Since H is normal $bH = Hb$ i.e. $bHb^{-1} = H$.

Note that $b_1Hb^{-1} = b \cdot h_2 \cdot H \cdot b^{-1}$ i.e. $\forall h_3 \in H$
 $\underbrace{b \cdot h_2 \cdot h_3 \cdot b^{-1}}_{\in H} \in H$ as H is normal.

Hence $a_1b_1Hb^{-1} = a_1H \overset{a_1h}{=} aH$

Since

In particular $a_1b_1Hb^{-1} = aH$ and so

$a^{-1} \cdot a_1b_1Hb^{-1} = H$ $\swarrow$ H is normal

Hence $a^{-1}a_1b_1H = Hb = bH$ and we

conclude that $a_1b_1H = abH$ as required.

" $\rightarrow$ " Assume now that the product operation is well defined i.e. $(aH) \cdot (bH) = abH$.

claim H is a normal subgroup of G .

Let $x \in aH$. We choose representatives

$x \in aH$ and $a^{-1} \in a^{-1}H$ we see that

~~aH~~ $(xH) \cdot (a^{-1}H) = H$ $\leftarrow$ since the operation is well defined.

then as $(aH)(a^{-1}H) = H$.

then $xa^{-1}H = H$ this means that

$xa^{-1} = h$ for some $h \in H$. consequently

$x = h \cdot a$

so every element $x \in aH$ $x = h \cdot a$ for some $h \in H$, then $aH = Ha$.

Corollary: Let H be a normal subgroup then $\{G/H = \{gH \mid g \in G\}$ + set of cosets, with the operation:

$$G/H \times G/H \longrightarrow G/H$$
$$(aH, bH) \longrightarrow abH \text{ forms a group.}$$

Proof:

• associativity: $a, b, c \in G$.

$$(aH) \cdot ((bH) \cdot (cH)) = (aH) \cdot (b \cdot cH) = (a \cdot (b \cdot c))H \\ = ((a \cdot b) \cdot c)H = (a \cdot b)H \cdot cH = ((aH) \cdot (bH)) \cdot cH.$$

• inverses: given $a \in G$ the inverse of aH is $a^{-1}H$.

• identity: $eH = H$.

This group has a name: it is called the factor group of G by H (or the quotient group).

e.g: $(\mathbb{Z}, +, 0) = G$ $H = (n\mathbb{Z}, +, 0)$

$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ is the factor group.

Main Goal: Let $(G, \cdot, e_G)$ be a group

there is a one to one correspondence:

- $N = \{ H \trianglelefteq G \mid G \text{ is normal} \}$
- $\text{Ker} = \{ S \leq G \mid S = \text{Ker}(\psi) \text{ for some } \psi: G \rightarrow K \text{ homomorphism} \}$

$$H \trianglelefteq G \longleftrightarrow \text{Ker} \quad \text{surjective}$$

Given $\psi: G \rightarrow K$ an homomorphism then $S = \text{Ker}(\psi)$ is a subgroup that is normal.

Does every normal subgroup come in this form? Let $H \trianglelefteq G$ can I find ψ an homomorphism whose kernel is H ?

$$\text{Let } \psi: G \longrightarrow G/H$$

$$g \longrightarrow gH$$

we just need to check that ψ is an homomorphism: $\psi(g_1 \cdot g_2) = g_1 \cdot g_2 H = (g_1 H) \cdot (g_2 H) = \psi(g_1) \cdot \psi(g_2)$ as required.

How does this look in our favorite example?

$$G = (\mathbb{Z}, +, 0) \quad H = (n\mathbb{Z}, +, 0)$$

Factor group is $G/H = \mathbb{Z}_n$.

$$\varphi: \mathbb{Z} \longrightarrow \mathbb{Z}_n$$

$$\mathbb{Z} \longrightarrow [\mathbb{Z}] \bmod n.$$

$$\text{Ker}(\varphi) = H!$$

THE FUNDAMENTAL HOMOMORPHISM THM:

Let $\varphi: G \longrightarrow G'$ be a group homomorphism with kernel $\neq H$ then $\varphi(G)$ is a group

and $\mu: G/H \longrightarrow \varphi(G)$

$gH \longrightarrow \varphi(g)$ is an isomorphism.

If $\gamma: G \longrightarrow G/H$

$g \longrightarrow gH$

then $\varphi(g) = \varphi \circ \gamma(g)$

$\forall g \in G.$

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & \varphi(G) \leq G' \\ \gamma \downarrow & \# & \uparrow \mu \\ & & \text{iso} \\ G/\text{Ker}(\varphi) & & \end{array}$$

Proof: We must verify that φ is a bijection and an homomorphism.

• bijection: injective: Assume $\varphi(g_1) = \varphi(g_2)$
 $\Rightarrow \varphi(g_1) \cdot \varphi(g_2^{-1}) = e_K \Rightarrow \varphi(g_1 \cdot g_2^{-1}) = e_K$
 $\Rightarrow g_1 g_2^{-1} \in \text{Ker}(\varphi) \Rightarrow g_1 \text{Ker}(\varphi) = g_2 \text{Ker}(\varphi)$.
i.e. $g_1 H = g_2 H$.

surjective: we restrict to $\varphi(G)$.

Given $g \in G$ $\varphi(gH) = \varphi(g)$.

homomorphism: We aim to show that

$$\mu((g_1 H) \cdot (g_2 H)) = \mu(g_1 H) \cdot \mu(g_2 H)$$

$$\mu((g_1 H) (g_2 H)) = \mu((g_1 g_2) H) = \varphi(g_1 g_2)$$

on the other hand $\mu(g_1 H) = \varphi(g_1)$ and
 $\mu(g_2 H) = \varphi(g_2)$ since φ is an homomorphism
so is μ .

• well defined! $\mu(gH) = \varphi(g)$

let $g_1 \in gH$ then $g_1 = g \cdot h$ then

$$\varphi(g_1) = \varphi(g) \cdot \varphi(h) = \varphi(g) \cdot e_K = \varphi(g)$$

• Place characterization of normal subgroups in the HWK.

Let's slow down and see a few examples.

We have seen:

① Factor groups.

Goal 1 If $H \leq G$ then

$$H \leq G \iff * : G/H \times G/H \rightarrow G/H$$
$$(g_1H, g_2H) \rightarrow g_1g_2H$$

is well defined

~~In p~~
~~where~~ In particular $(G/H, *, H)$ is a group.

Goal 2 Let $\varphi : (G, \circ_G, e_G) \rightarrow (G', \circ_{G'}, e_{G'})$

be a group homomorphism. Then:

$\text{Ker}(\varphi) = \{g \in G \mid \varphi(g) = e_{G'}\}$ is a normal subgroup of G .

Goal 3 Let $(G, \circ_G, e_G)$ be a group.

$$\mathcal{H} = \{H \leq G \mid H \trianglelefteq G\}$$

$$\mathcal{K} = \{K \leq G \mid K = \text{Ker}(\varphi) \text{ where } \varphi : G \rightarrow (G', \circ_{G'}, e_{G'})$$

is any group homomorphism

There is a one to one correspondence between $\mathcal{H}$ and $\mathcal{K}$.

But how does this look on some examples?

• $G = (\mathbb{Z}, +, 0)$ $H = (n\mathbb{Z}, +, 0)$

→ H is normal in G because G is abelian.

$$G/H = \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$$

cosets are: $0+n\mathbb{Z} = n\mathbb{Z}$, $1+n\mathbb{Z}$, ..., $(n-1)+n\mathbb{Z}$
 $[0]_n$ $[1]_n$... $[n-1]_{\text{mod } n}$.

• ~~Can we see H as the kernel of an homo~~

How does the operation look like?

$$[z_1]_n + [z_2]_n = [z_1 + z_2]_n.$$

$$(z_1 + n\mathbb{Z}, z_2 + n\mathbb{Z}) \rightarrow (z_1 + z_2) + n\mathbb{Z}$$

Can we see H as the kernel of an ~~homomorphism~~ homomorphism?

$$\varphi: \mathbb{Z} \longrightarrow \mathbb{Z}_n$$

$$z \longrightarrow [z]_n.$$

← the midterm
 φ is an hom.

$$\text{Ker}(\varphi) = \{z \in \mathbb{Z} \mid \varphi(z) = [0]_n\}$$

$$= \{z \in \mathbb{Z} \mid [z]_n = [0]_n\} = n\mathbb{Z}$$

Example 2: consider $(\mathbb{C}^X, \cdot, 1)$

$$|\cdot|: (\mathbb{C}^X, \cdot, 1) \longrightarrow (\mathbb{R}^+, \cdot, 1)$$

$$z \longmapsto |z|$$

Note that $|\cdot|$ is an homomorphism

since: ~~$(\mathbb{C}^X, \cdot, 1)$~~ ~~$|\cdot|$~~

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

Recall: $\mathbb{C}^X \times \mathbb{C}^X \longrightarrow \mathbb{C}^X$
 $(|z_1|e^{i\theta_1}, |z_2|e^{i\theta_2}) \longrightarrow |z_1 \cdot z_2|e^{i(\theta_1 + \theta_2)}$

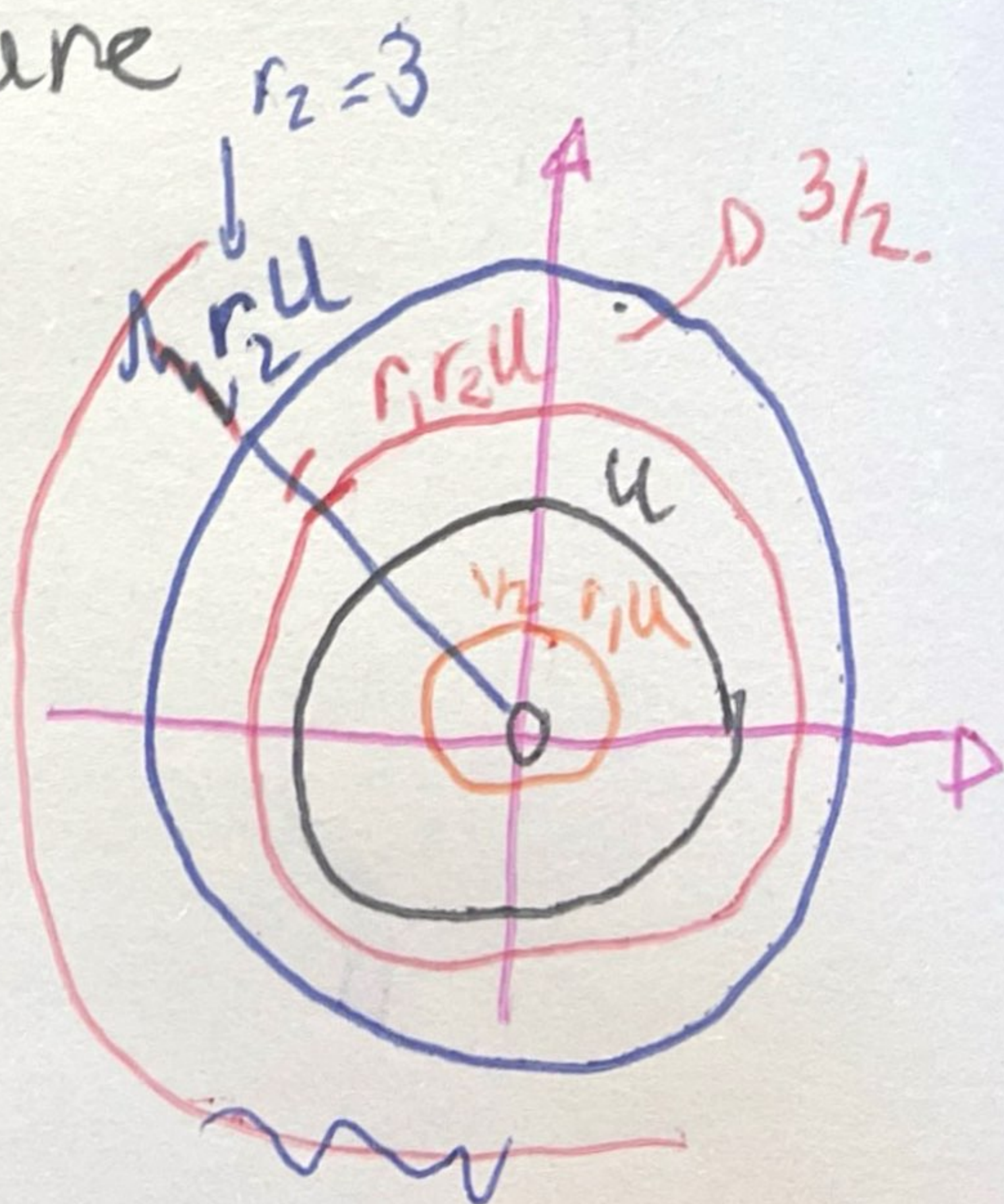
$$\ker(|\cdot|) = \{z \in \mathbb{C}^X \mid |z| = 1\} = U. \quad \begin{array}{l} \text{is} \\ \text{normal} \\ \text{in } (\mathbb{C}^X, \cdot, 1) \end{array}$$

How does $\mathbb{C}^X/U$ look like?

$\rightarrow$ circles of radius r are the cosets!

multiplying z of them

$$(r_1 u) \cdot (r_2 u) = (r_1 \cdot r_2) u.$$



Example 3: (From Calculus)

$$D = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is differentiable}\}$$

$$F = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is a function}\}.$$

$(F, +, \text{id}_{\mathbb{R}})$ is a group.

$(D, +, \text{id}_{\mathbb{R}})$ is a group.

$$\begin{aligned} \psi: D &\longrightarrow F && \text{the derivative.} \\ f &\longrightarrow f' \end{aligned}$$

ψ is an homomorphism $\psi(f_1 + f_2) = \psi(f_1) + \psi(f_2)$

$$(f_1 + f_2)' = f_1' + f_2' = \psi(f_1) + \psi(f_2)$$

$$\text{Ker}(\psi) = \{f \in D \mid f' = \text{id}_{\mathbb{R}}\} = \{c \in \mathbb{R}\}$$

i.e. the constant functions: $(c)' = 0$.

What if I try to find a coset of $\text{Ker}(\psi)$.

How does it look like? $\forall$ for $f \in D$

$$f + \text{Ker}(\psi) = f + c \text{ such that } c \in \mathbb{R}.$$

$$\begin{aligned} \psi(f + c) &= f + \text{Ker}(\psi) = \{g \in D \mid \psi(g) = \psi(f)\} \\ &= \{g \in D \mid g' = f'\}. \end{aligned}$$

Example 4: S_n : permutation group of n -elements
 A_n : alternating group.

Claim: A_n is a normal subgroup of S_n .

Consider $\varphi : S_n \longrightarrow \{1, -1\} = \mathbb{C}_2$
 $\sigma \longmapsto \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$

$\star$	1	-1
1	1	-1
-1	-1	1

$\leftarrow$ group with identity 1.

is an homomorphism

Given $\sigma, \tau \in S_n$ ~~# of trans~~

of transpositions of $\sigma \cdot \tau = \# \text{ of transpositions of } \sigma + \# \text{ of transpositions of } \tau.$

- σ is even σ uses $2k$ trans.

τ is even τ uses $2m+1$

$\sigma \cdot \tau$ uses $2k + 2m+1$

$\varphi(\sigma \cdot \tau) = -1 = \underbrace{\varphi(\sigma)}_1 \star \underbrace{\varphi(\tau)}_{-1}$ and so on.

$$\text{Ker}(\chi) = \{\sigma \in S_n \mid \chi(\sigma) = 1\} = A_n.$$

$$|S_n| = n! \quad |A_n| = \frac{n!}{2}.$$

$$|S_n/A_n| = 2.$$

σ odd.

$S_n/A_n = \{A_n, \sigma A_n\}$ then the group operation is:

$$(\text{even})(\text{even}) = \text{even} \quad (\text{odd})(\text{even}) = \text{odd}$$

$$(\text{even})(\text{odd}) = \text{odd} \quad (\text{odd})(\text{odd}) = \text{even}$$

table:

	A_n	σA_n
A_n	A_n	σA_n
σA_n	σA_n	A_n

Example 5: $G = (GL_n(\mathbb{R}), \cdot, I)$ $G' = (\mathbb{R}^x, \cdot, 1)$

$$\det: GL_n(\mathbb{R}) \rightarrow \mathbb{R}^x$$

$$A \mapsto \det(A)$$

$\det$ is an homomorphism:

$$\det(A \cdot B) = \det(A) \cdot \det(B)$$

$$\text{Ker}(\det) = \{ A \in \text{GL}_n(\mathbb{R}) \mid \det(A) = 1 \}$$

↳ special group. $\text{SL}_n(\mathbb{R})$
linear.

How do cosets look like?

$$\text{GL}_n(\mathbb{R}) / \text{Ker}(\det)$$

$$B \cdot \text{SL}_n(\mathbb{R}) = \{ B \cdot A \mid \det(A) = 1 \}$$

The operation is:

$$(B_1 \text{SL}_n(\mathbb{R}), B_2 \text{SL}_n(\mathbb{R})) \longrightarrow B_1 \cdot B_2 \text{SL}_n(\mathbb{R})$$

Theorem: Let $H \times G$ be the direct product of H and G groups.

Let $K_H = \{ (h, e_G) \mid h \in H \}$ then K_H is normal in $H \times G$, similarly $K_G = \{ (e_H, g) \mid g \in G \}$ is normal in $H \times G$.

Proof: Let $(h_1, g_1) \in H \times G$

We want to show $(h_1, g_1) \cdot K_H = K_H (h_1, g_1)$

$$(h_1, g_1) \cdot (h', e_G) = (h_1 \cdot h', g_1)$$

$$= (h_1 \cdot h' \cdot h_1^{-1} \cdot h_1, g_1)$$

$$= (h_1 \cdot h' \cdot h_1^{-1}, e_G) \cdot (h_1, g_1) \in K_H (h_1, g_1)$$

$$H \times G / K_H \cong G.$$

↳ there is a natural projection map:

$$\begin{aligned} \pi: H \times G &\longrightarrow G \\ (h, g) &\longrightarrow g. \end{aligned}$$

π is an hom and $\text{Ker}(\pi) = K_H$.

$$\begin{aligned} \tilde{\pi}: H \times G / K_H &\longrightarrow G \quad \text{is an isomorphism.} \\ (h, g) K_H &\longrightarrow g. \end{aligned}$$

Example: $\mathbb{Z}_4 \times \mathbb{Z}_6 / \langle (0, 2) \rangle$

$$\langle (0, 2) \rangle = \{(0, 0), (0, 2), (0, 4)\} \subseteq H. \quad \rightarrow \text{order 3.}$$

$$\mathbb{Z}_4 \times \mathbb{Z}_6 / H$$

$(z_1, w) + H$ the $\mathbb{Z}_4$ part is left alone while the second coordinate is quotient by $\langle 2 \rangle$
 $\downarrow$
 of order 3.

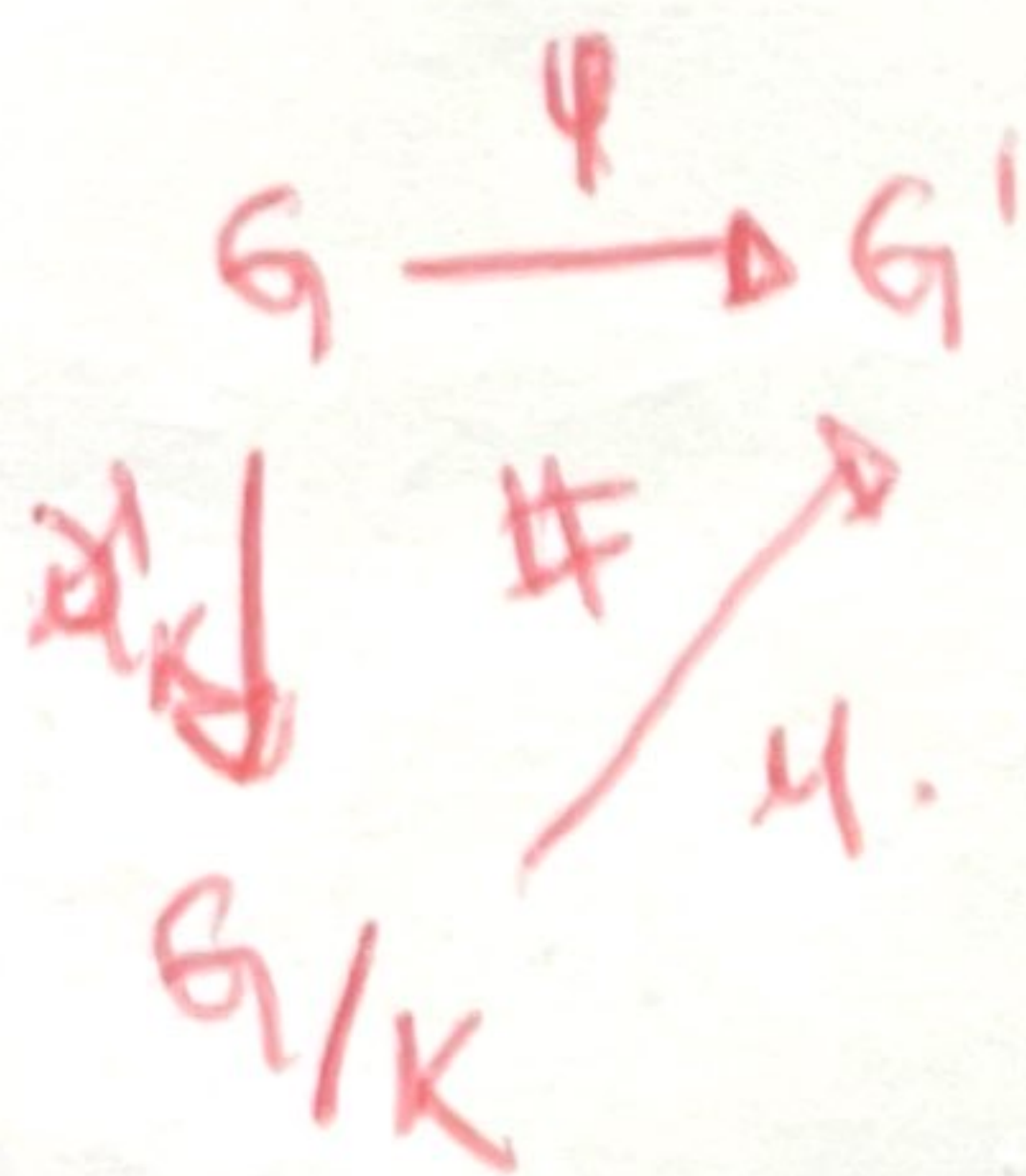
$$\mathbb{Z}_6 / H \cong \mathbb{Z}_2.$$

Factor group of a cyclic group is cyclic.

Isomorphism Theorems:

THM 1: (First isomorphism theorem) Let $\varphi: G \rightarrow G'$ be a homomorphism with kernel K and let $\pi_K: G \rightarrow G/K$ be the canonical homomorphism. There is a unique isomorphism

$$\begin{aligned} \psi: G/K &\rightarrow G' \\ aK &\mapsto \varphi(a) \end{aligned}$$



Lemma: Let N be a normal subgroup of a group G and let $\pi: G \rightarrow G/N$ be the canonical homomorphism.

$$\text{Let } \mathcal{A} = \{ H \leq G \mid N \subseteq H \}$$

$$\mathcal{B} = \{ \bar{H} \leq G/N \}$$

consider the map: $\varphi: \eta \rightarrow R$

$$H \rightarrow H/N = \tilde{H}$$

is one-to-one.

Proof: Assume

We need a previous statement:

Lemma 1: let $\varphi: G \rightarrow G'$ be a group morphism

If N is a normal subgroup of G , then $\varphi[N]$ is a normal subgroup of $\varphi[G]$. Also if N' is a normal subgroup of $\varphi[G]$ then $\varphi^{-1}(N')$ is a normal subgroup of G .

Proof: $\rightarrow$ Assume N is a normal subgroup of G , then $\varphi[N]$ is a subgroup of $\varphi[G]$.

Now we check normality, let $g \in G$

$$\varphi(g) \in \varphi[G].$$

We aim to show $\varphi(g) \varphi(N) \varphi(g)^{-1} = \varphi(N)$

$\subseteq$ let $n \in N$, since φ is an homomorphism

$$\varphi(g) \cdot \varphi(n) \varphi(g)^{-1} = \varphi(g \cdot n g^{-1}) = \varphi(n_2) \in N.$$

$\underbrace{\hspace{1cm}}_{= n_2 \in N}$

$\supseteq$ let $n \in N$ then consider $g^{-1} n g$ by a similar argument to the one above

Since N is normal, $g^{-1}ng \in N$
 ~~$\varphi(g^{-1}ng) = \varphi(n)$~~

$$\varphi(n) = \varphi(g^{-1}ng) = \varphi(g^{-1}) \cdot \varphi(n) \cdot \varphi(g)$$

thus $\varphi(n) \in \varphi(g) \varphi(N) \varphi(g)^{-1}$

For the converse note that φ^{-1} is an homomorphism. (add details φ^{-1} might not be defined) (see claim *)

Lemma: Let N be a normal subgroup of G and

let $\pi: G \rightarrow G/N$ be the canonical homo-

morphism. the map $\varphi: H \rightarrow \tilde{H}$ is one to one

$$H \rightarrow \tilde{H} = H/N$$

$$H = \{h \in G \mid N \subseteq h\}$$

$$\tilde{H} = \{h \in G/N\}$$

Proof: Assume $H \trianglelefteq G$ such that $N \subseteq H$.

Then by the previous statement $\pi(H)$ is also normal in G/N i.e. H/N is a normal subgroup of G/N , so the map is one to one makes sense.

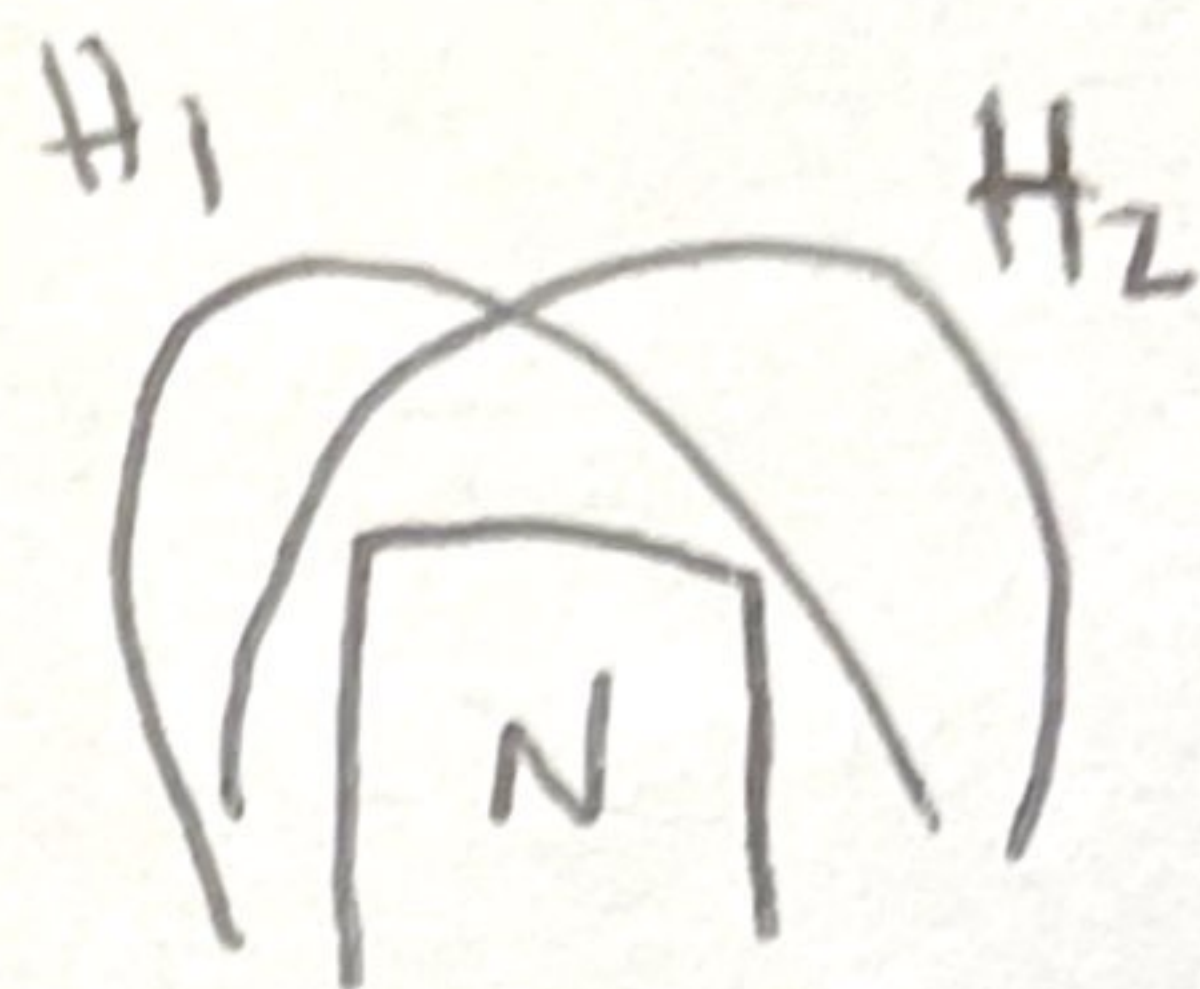
or injectivity: Assume that $\varphi(h_1) = \varphi(h_2)$

Then $H_1/N = H_2/N$, we aim to show that $H_1 = H_2$.

since $H_1/N = H_2/N$ then $\forall h_1 \in H_1$

$h_1 N = h_2 N$ for some $h_2 \in H_2$

then $h_2^{-1} \cdot h_1 = n \in N$.



In particular $h_1 = h_2 \cdot n \in H_2$ so $H_1 \subseteq H_2$

a similar argument shows that $H_2 \subseteq H_1$.

Let $\hat{H}$ be a subgroup of G/N .

Let $H = \{x \in G \mid \varphi(x) \in \hat{H}\}$.

(this proof works for the general case)

* Claim: H is a normal subgroup of G .

We first verify that it is a subgroup.

①. $e_G \in H$ since $\varphi(e_G) = e_G N \in \hat{H} \trianglelefteq G/N$.

②. closed under operation: let $g_1, g_2 \in H$

we want to show $g_1 \circ g_2 \in H$.

since φ is an homomorphism then

$$\varphi(g_1 \circ g_2) = \underbrace{\varphi(g_1)}_{\in \hat{H}} \cdot \underbrace{\varphi(g_2)}_{\in \hat{H}} \in \hat{H}.$$

③ inverses: Assume $g \in H$ then $\varphi(g) \in \hat{H}$.

~~since $\hat{H}$ is a subgroup $\exists h \in \hat{H}$ s.t.~~

$$\varphi(g) \cdot h =$$

let $g \in G$ and assume $g \in H$ i.e. $\phi(g) \in \hat{H}$.
we want to show that $g^{-1} \in H$.

since ϕ is an homomorphism $\phi(g) \cdot \phi(g^{-1}) = \phi(g \cdot g^{-1}) = \phi(e_G)$

then $\phi(g) \cdot \phi(g^{-1}) = e_{G_1} = e_{G \cap N}$.

thus $\phi(g^{-1})$ is an inverse in the image,
since inverses are unique and $\hat{H}$ is
a subgroup then $\phi(g^{-1}) \in \hat{H}$.

Now we verify that H is normal:

we want to show that $\forall g \in G$
 $gHg^{-1} = H$.

Then let $h \in H$ i.e. $\phi(h) \in \hat{H}$ we
want to prove that $ghg^{-1} \in H$ i.e.
 $\phi(ghg^{-1}) \in \hat{H}$.

since ϕ is an homomorphism

$$\phi(g) \phi(h) \phi(g)^{-1} = \phi(ghg^{-1})$$

then since $\hat{H}$ is normal

$\phi(g) \phi(h) \phi(g)^{-1} \in \hat{H}$ i.e. $\phi(ghg^{-1}) \in \hat{H}$
so $ghg^{-1} \in H$, as required.

then the map φ is surjective: Given $\hat{H} \trianglelefteq G/N$ then $\varphi^{-1}(\hat{H})$ is a normal subgroup that contains N .

If H and N are subgroups of G then we

let $HN = \{hn \mid h \in H, n \in N\}$

We define the JOIN $H \vee N$ of H and N as the intersection of all subgroups of G that contain HN .

then $H \vee N$ is the smallest subgroup that contains HN .

Warning: In general HN might not be a subgroup! that's why we go to $H \vee N$.

But we have the following lemma:

Lemma: If $N \trianglelefteq G$ and $H \leq G$ then

①. $H \vee N = HN = NH$.

②. if H is also normal in G then HN is also normal in G .

Proofs $N \trianglelefteq G$ and $H \leq G$

For ① we show that NH is a subgroup:

$$\bullet e_G = \underbrace{e_G}_{\in N} \cdot \underbrace{e_G}_{\in H} \in NH.$$

• closed under $\cdot$: let $n_1, n_2 \in N$ and $h_1, h_2 \in H$

we want to show $(n_1 h_1) \cdot (n_2 h_2) \in NH$

i.e. it can be expressed as a product of $n \in N$ and $h \in H$.

$$(n_1 h_1) \cdot (n_2 h_2) = n_1 (h_1 n_2) h_2 *$$

Since N is normal $h_1 N = N h_1$ i.e. $h_1 n_2 = n_3 h_1$

for $n_3 \in N$ then $* = n_1 (n_3 h_1) \cdot h_2$

$$= n_1 \cdot n_3 \cdot (h_1 h_2) \in NH.$$

closed under ^{inverse} product: let $nh \in NH$

$(nh)^{-1} = h^{-1} \cdot n^{-1}$ we want to show $\in NH$.

$n^{-1} \in N$ since $N < G$

then $h^{-1} \cdot n^{-1} = \underbrace{n_3}_{\text{so we expressed it as a product.}} h^{-1}$ as $N \trianglelefteq G$
 $h^{-1} \in H$

(2) Now suppose that H is also normal.

We aim to show $NH \trianglelefteq G$ let $g \in G$

it is sufficient to show $g \cdot (NH) = (NH)g$

i.e. the cosets are the same.

let $n \in N$ and $h \in H$ then $g(n \cdot h) =$

$(g \cdot n) \cdot h$ since $N \trianglelefteq G$ $g \cdot n = n_3 \cdot g$
for some $n_3 \in N$.

so $(g \cdot n) \cdot h = (n_3 \cdot g) \cdot h = n_3 \cdot (gh)$ since H is normal
 $= n_3 \cdot h_4 \cdot g$.

$gh = h_4 \cdot g$
for $h_4 \in H$

thus $gNh \subseteq Nhg$ a similar

argument shows the other containment.

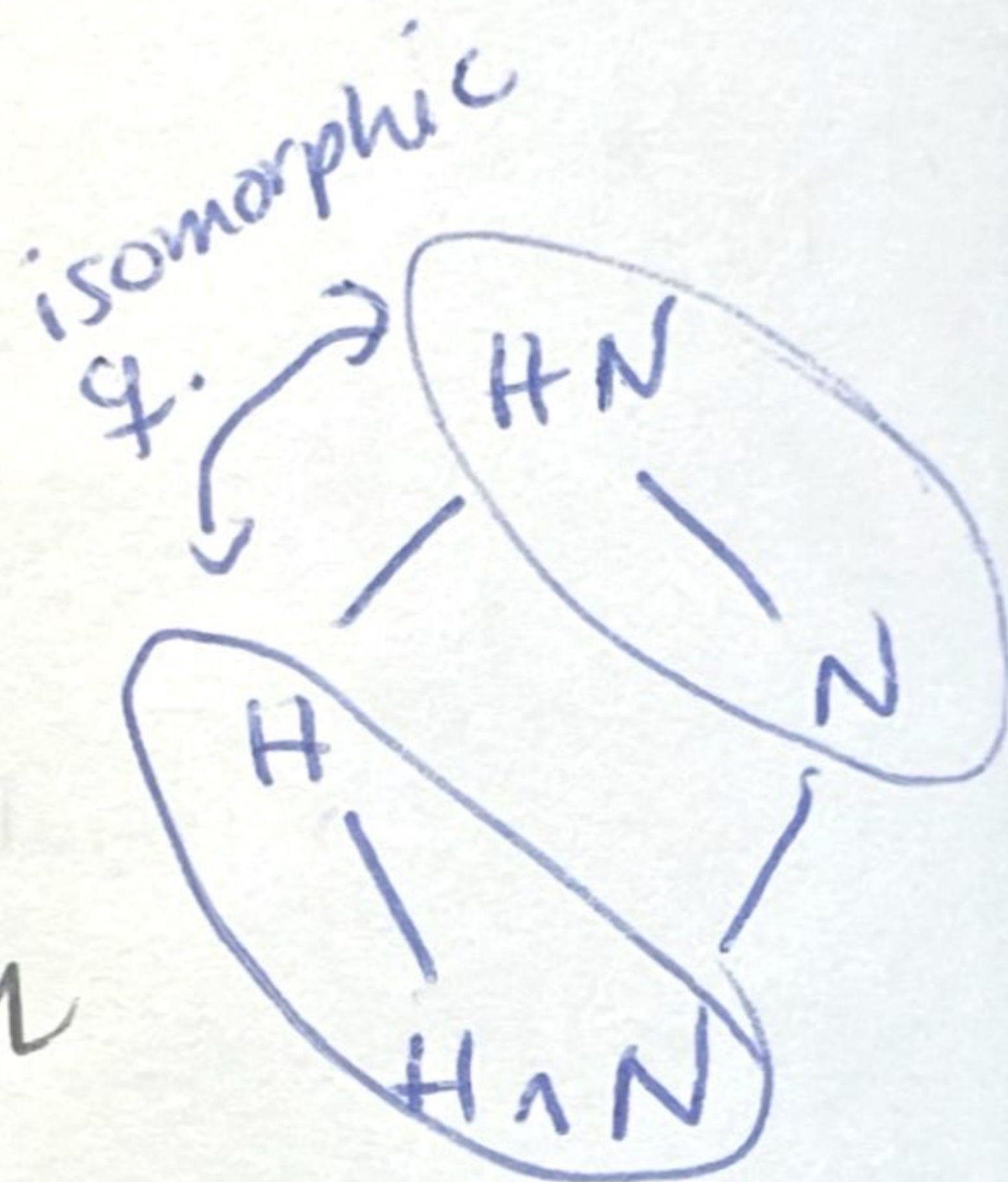
THEOREM: (The 2nd ISOMORPHISM THM)

Let $H \leq G$ and $N \leq G$ then ~~HN/N~~

$$HN/N \cong H/H \cap N.$$

Proof: let

$\varphi: G \rightarrow G/N$ be
the canonical projection
map.



we might consider the restriction of
this map to h i.e. $\varphi: H \rightarrow G/N$
 $h \rightarrow hN.$

By definition, the image of φ is HN/N
the restriction of an homomorphism is
still an homomorphism. so we have the
map $\varphi|_H: H \rightarrow HN/N$ that is
a surjective
homomorphism.

What is its kernel?

$$\text{Ker}(\psi|_H) = \{h \in H \mid hN = N\}$$

$hN = N$ iff $h \in N$ then

$$\text{Ker}(\psi|_H) = H \cap N.$$

By the 1st iso,

$$H / H \cap N \cong \cancel{HN} / N$$

Example: $G = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ $H = \mathbb{Z} \times \mathbb{Z} \times \{0\}$

$$N = \{0\} \times \mathbb{Z} \times \mathbb{Z} \quad HN = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$$

$$H \cap N = \{0\} \times \mathbb{Z} \times \{0\} \quad HN / N \cong \mathbb{Z}$$

$$H / H \cap N \cong \mathbb{Z}.$$

[THIRD ISOMORPHISM THM] Let H and K be normal subgroups with $K \leq H$.

$$\text{Then } G/H \cong G/K / H/K$$

Proof:

$$\text{Let } \varphi: G/K \longrightarrow G/H.$$

$$gK \longrightarrow gH.$$

$$\left[\begin{array}{c} G \\ \triangle \\ H \\ \triangle \\ K \end{array} \right]_+$$

claim: φ is ~~an~~ ~~to~~ well defined,

$$\text{let } g_1K = g_2K \Rightarrow g_2^{-1}g_1 \in K \leq H$$

$$\text{so } g_2^{-1}g_1 \in H \text{ and } g_2H = g_1H.$$

claim: φ is an homomorphism.

$$\varphi(g_1K)(g_2K) = \varphi(g_1g_2K) = g_1g_2H$$

$$= (g_1H) \cdot (g_2H) \in G/H.$$

$$\underbrace{\varphi(g_1K)} \quad \underbrace{\varphi(g_2K)}$$

Note that φ is surjective by def.

By the first iso theorem:

$$(G/K)/\ker(\varphi) \cong G/H.$$

$$\ker(\varphi) = \{gK \mid gH = H\}$$

$$gH = H \text{ iff } g \in H$$

$$\ker(\varphi) = \{gK \mid g \in H\} = H/K$$

as required.